Chapter 3: Vectors: Problems with Solutions

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1. The polar coordinates of a point are $r = 5.50$ m and $\theta = 240°$. What are the Cartesian coordinates of this point?

W

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

Then,

$$x = r\cos\theta = (5.50\text{ m})\cos 240°$$

$$= (5.50\text{ m})(-0.5) = \boxed{-2.75\text{ m}}$$

$$y = r\sin\theta = (5.50\text{ m})\sin 240°$$

$$= (5.50\text{ m})(-0.866) = \boxed{-4.76\text{ m}}$$

3. Two points in the *xy* plane have Cartesian coordinates (2.00, −4.00) m and (−3.00, 3.00) m. Determine (a) the distance between these points and (b) their polar coordinates.

(a) The distance between the points is given by

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(2.00 - [-3.00])^2 + (-4.00 - 3.00)^2}$$

$$d = \sqrt{25.0 + 49.0} = \boxed{8.60 \text{ m}}$$

(b) To find the polar coordinates of each point, we measure the radial distance to that point and the angle it makes with the +*x* axis:

$$r_1 = \sqrt{(2.00)^2 + (-4.00)^2} = \sqrt{20.0} = \boxed{4.47 \text{ m}}$$

$$\theta_1 = \tan^{-1}\left(-\frac{4.00}{2.00}\right) = \boxed{-63.4^\circ}$$

$$r_2 = \sqrt{(-3.00)^2 + (3.00)^2} = \sqrt{18.0} = \boxed{4.24 \text{ m}}$$

$\theta_2 =$ $\boxed{135^\circ}$ measured from the +*x* axis.

15. A vector has an x component of -25.0 units and a y component of 40.0 units. Find the magnitude and direction of this vector.

W

$$A = \sqrt{A_x^2 + A_y^2} = \sqrt{(-25.0)^2 + (40.0)^2}$$

$$= \boxed{47.2 \text{ units}}$$

$$\phi = \tan^{-1}\left(\frac{A_y}{|A_x|}\right) = \tan^{-1}\left(\frac{40.0}{25.0}\right) = \tan^{-1}(1.60) = 58.0°$$

The diagram shows that the angle from the $+x$ axis can be found by subtracting from 180°: $\theta = 180° - 58° = \boxed{122°}$

23. M Consider the two vectors $\vec{\mathbf{A}} = 3\hat{\mathbf{i}} - 2\hat{\mathbf{j}}$ and $\vec{\mathbf{B}} = -\hat{\mathbf{i}} - 4\hat{\mathbf{j}}$. Calculate (a) $\vec{\mathbf{A}} + \vec{\mathbf{B}}$, (b) $\vec{\mathbf{A}} - \vec{\mathbf{B}}$, (c) $|\vec{\mathbf{A}} + \vec{\mathbf{B}}|$, (d) $|\vec{\mathbf{A}} - \vec{\mathbf{B}}|$, and (e) the directions of $\vec{\mathbf{A}} + \vec{\mathbf{B}}$ and $\vec{\mathbf{A}} - \vec{\mathbf{B}}$.

(a) $$\left(\vec{\mathbf{A}} + \vec{\mathbf{B}}\right) = \left(3\hat{\mathbf{i}} - 2\hat{\mathbf{j}}\right) + \left(-\hat{\mathbf{i}} - 4\hat{\mathbf{j}}\right) = \boxed{2\hat{\mathbf{i}} - 6\hat{\mathbf{j}}}$$

(b) $$\left(\vec{\mathbf{A}} - \vec{\mathbf{B}}\right) = \left(3\hat{\mathbf{i}} - 2\hat{\mathbf{j}}\right) - \left(-\hat{\mathbf{i}} - 4\hat{\mathbf{j}}\right) = \boxed{4\hat{\mathbf{i}} + 2\hat{\mathbf{j}}}$$

(c) $$\left|\vec{\mathbf{A}} + \vec{\mathbf{B}}\right| = \sqrt{2^2 + 6^2} = \boxed{6.32}$$

(d) $$\left|\vec{\mathbf{A}} - \vec{\mathbf{B}}\right| = \sqrt{4^2 + 2^2} = \boxed{4.47}$$

(e) $$\theta_{|\mathbf{A}+\mathbf{B}|} = \tan^{-1}\left(-\frac{6}{2}\right) = -71.6^\circ = \boxed{288^\circ}$$

$$\theta_{|\mathbf{A}-\mathbf{B}|} = \tan^{-1}\left(\frac{2}{4}\right) = \boxed{26.6^\circ}$$

32. Vector $\vec{A}$ has x and y components of -8.70 cm and 15.0 cm, respectively; vector $\vec{B}$ has x and y components of 13.2 cm and -6.60 cm, respectively. If $\vec{A} - \vec{B} + 3\vec{C} = 0$, what are the components of $\vec{C}$?

We are given $\vec{A} = -8.70\hat{i} + 15.0\hat{j}$, and $\vec{B} = 13.2\hat{i} - 6.60\hat{j}$, and $\vec{A} - \vec{B} + 3\vec{C} = 0$. Solving for $\vec{C}$ gives

$$3\vec{C} = \vec{B} - \vec{A} = 21.9\hat{i} - 21.6\hat{j}$$

$$\vec{C} = 7.30\hat{i} - 7.20\hat{j} \text{ or } \quad C_x = \boxed{7.30 \text{ cm}}; C_y = \boxed{-7.20 \text{ cm}}$$

34. Vector $\vec{\mathbf{B}}$ has x, y, and z components of 4.00, 6.00, and 3.00 units, respectively. Calculate (a) the magnitude of $\vec{\mathbf{B}}$ and (b) the angle that $\vec{\mathbf{B}}$ makes with each coordinate axis.

We are given $\vec{\mathbf{B}} = B_x\hat{\mathbf{i}} + B_y\hat{\mathbf{j}} + B_z\hat{\mathbf{k}} = 4.00\hat{\mathbf{i}} + 6.00\hat{\mathbf{j}} + 3.00\hat{\mathbf{k}}$. The magnitude of the vector is therefore

$$\left|\vec{\mathbf{B}}\right| = \sqrt{4.00^2 + 6.00^2 + 3.00^2} = \boxed{7.81}$$

And the angle of the vector with the three coordinate axes is

$$\alpha = \cos^{-1}\left(\frac{4.00}{7.81}\right) = \boxed{59.2^\circ}$$ is the angle with the x axis

$$\beta = \cos^{-1}\left(\frac{6.00}{7.81}\right) = \boxed{39.8^\circ}$$ is the angle with the y axis

$$\gamma = \cos^{-1}\left(\frac{3.00}{7.81}\right) = \boxed{67.4^\circ}$$ is the angle with the z axis

37. (a) Taking $\vec{\mathbf{A}} = (6.00\hat{\mathbf{i}} - 8.00\hat{\mathbf{j}})$ units, $\vec{\mathbf{B}} = (-8.00\hat{\mathbf{i}} + 3.00\hat{\mathbf{j}})$ units, and $\vec{\mathbf{C}} = (26.0\hat{\mathbf{i}} + 19.0\hat{\mathbf{j}})$ units, determine a and b such that $a\vec{\mathbf{A}} + b\vec{\mathbf{B}} + \vec{\mathbf{C}} = 0$. (b) A student has learned that a single equation cannot be solved to determine values for more than one unknown in it. How would you explain to him that both a and b can be determined from the single equation used in part (a)?

(a) Taking components along $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$, we get two equations:

$$6.00a - 8.00b + 26.0 = 0$$

and

$$-8.00a + 3.00b + 19.0 = 0$$

Substituting $a = 1.33b - 4.33$ into the second equation, we find

$$-8(1.33b - 4.33) + 3b + 19 = 0 \rightarrow 7.67b = 53.67 \rightarrow b = 7.00$$

and so $a = 1.33(7.00) - 4.33 = 5.00$.

Thus $\boxed{a = 5.00,\ b = 7.00}$. Therefore, $5.00\vec{\mathbf{A}} + 7.00\vec{\mathbf{B}} + \vec{\mathbf{C}} = 0$.

(b) In order for vectors to be equal, all of their components must be equal. A vector equation contains more information than a scalar equation, as each component gives us one equation.

38. Three displacement vectors of a croquet ball are shown in Figure P3.38, where $|\vec{A}| = 20.0$ units, $|\vec{B}| = 40.0$ units, and $|\vec{C}| = 30.0$ units. Find (a) the resultant in unit-vector notation and (b) the magnitude and direction of the resultant displacement.

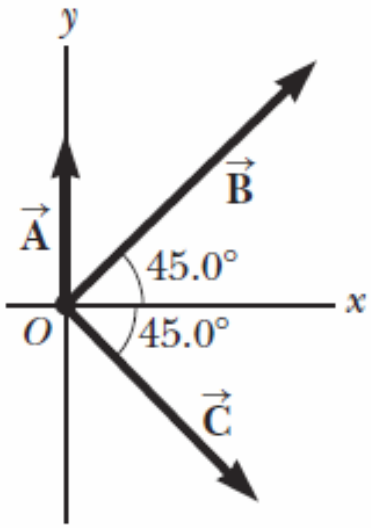
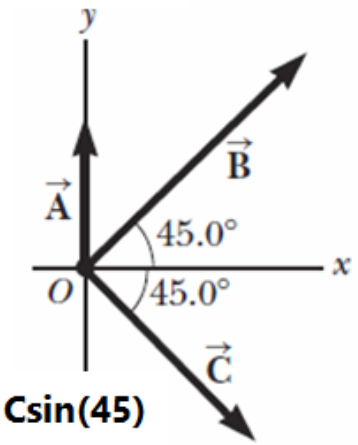


Figure P3.38



Bsin(45)

Bcos(45) + Ccos(45)

Csin(45)

Figure P3.38

$$\vec{R} = R_x i + R_y j$$

$$R_x = B\cos 45 + C\cos 45 - 0$$

$$= 40\cos 45 + 30\cos 45 = 49.5\ units$$

$$R_y = B\sin 45 + A - C\sin 45 =$$

$$= 40\sin 45 + 20 - 30\sin 45 = 27.1\ units$$

so $\vec{R} = \boxed{49.5\hat{i} + 27.1\hat{j}}$

(b) $|\vec{R}| = \sqrt{(49.5)^2 + (27.1)^2} = \boxed{56.4}$

$$\theta = \tan^{-1}\left(\frac{R_y}{R_x}\right) = \tan^{-1}\left(\frac{27.1}{49.5}\right) = \boxed{28.7°}$$

66. Vectors $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$ have equal magnitudes of 5.00. The sum of $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$ is the vector $6.00\hat{\mathbf{j}}$. Determine the angle between $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$.

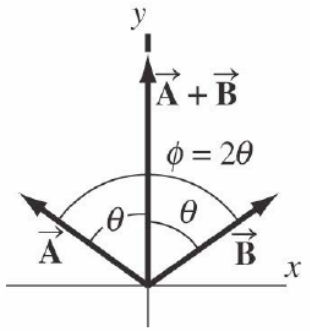
Since

$$\vec{\mathbf{A}} + \vec{\mathbf{B}} = 6.00\hat{\mathbf{j}},$$

we have

$$(A_x + B_x)\hat{\mathbf{i}} + (A_y + B_y)\hat{\mathbf{j}} = 0\hat{\mathbf{i}} + 6.00\hat{\mathbf{j}}$$

giving $A_x + B_x = 0 \rightarrow A_x = -B_x$.



ANS. FIG. P3.66

Because the vectors have the same magnitude and x components of equal magnitude but of opposite sign, the vectors are reflections of each other in the y axis, as shown in the diagram. Therefore, the two vectors have the same y components:

$$A_y = B_y = (1/2)(6.00) = 3.00$$

Defining θ as the angle between either $\vec{\mathbf{A}}$ or $\vec{\mathbf{B}}$ and the y axis, it is seen that

$$\cos\theta = \frac{A_y}{A} = \frac{B_y}{B} = \frac{3.00}{5.00} = 0.600 \rightarrow \theta = 53.1°$$

The angle between $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$ is then $\boxed{\phi = 2\theta = 106°}$.

Chapter 7.3: Scalar Product

8. Vector $\vec{A}$ has a magnitude of 5.00 units, and vector $\vec{B}$ has a magnitude of 9.00 units. The two vectors make an angle of 50.0° with each other. Find $\vec{A} \cdot \vec{B}$.

$A = 5.00;\ B = 9.00;\ \theta = 50.0°$

$$\vec{A} \cdot \vec{B} = AB\cos\theta = (5.00)(9.00)\cos 50.0° = \boxed{28.9}$$

9. For $\vec{A} = 3\hat{i} + \hat{j} - \hat{k}$, $\vec{B} = -\hat{i} + 2\hat{j} + 5\hat{k}$, and $\vec{C} = 2\hat{j} - 3\hat{k}$, find $\vec{C} \cdot (\vec{A} - \vec{B})$.

$$\vec{A} - \vec{B} = \left(3.00\hat{i} + \hat{j} - \hat{k}\right) - \left(-\hat{i} + 2.00\hat{j} + 5.00\hat{k}\right) = 4.00\hat{i} - \hat{j} - 6.00\hat{k}$$

$$\vec{C} \cdot \left(\vec{A} - \vec{B}\right) = \left(2.00\hat{j} - 3.00\hat{k}\right) \cdot \left(4.00\hat{i} - \hat{j} - 6.00\hat{k}\right) = 0 + (-2.00) + (+18.0)$$

$$= \boxed{16.0}$$

10. Find the scalar product of the vectors in Figure P7.10.

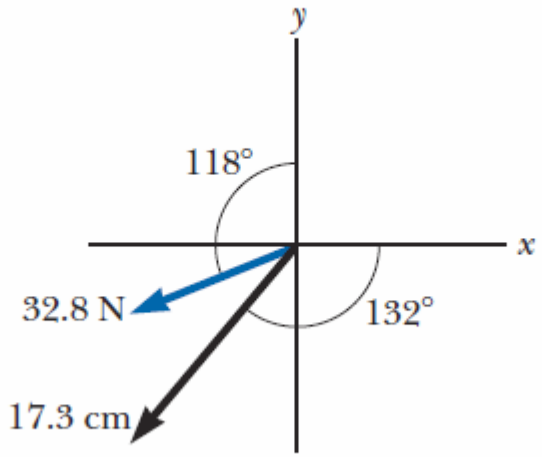


Figure P7.10

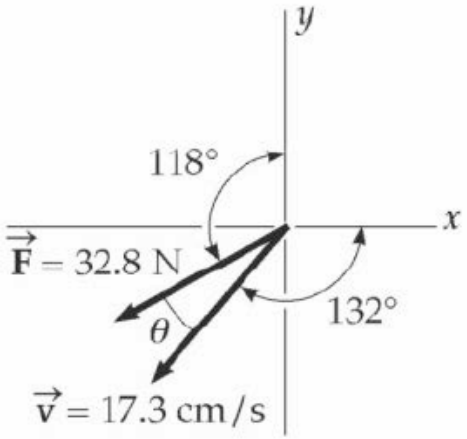
We must first find the angle between the two vectors. It is

$$\theta = (360^\circ - 132^\circ) - (118^\circ + 90.0^\circ)$$
$$= 20.0^\circ$$

Then

$$\vec{F} \cdot \vec{r} = Fr\cos\theta$$
$$= (32.8\ \text{N})(0.173\ \text{m})\cos 20.0^\circ$$

or $$\vec{F} \cdot \vec{r} = 5.33\ \text{N} \cdot \text{m} = \boxed{5.33\ \text{J}}$$



ANS. FIG. P7.10

12. Using the definition of the scalar product, find the angles between (a) $\vec{\mathbf{A}} = 3\hat{\mathbf{i}} - 2\hat{\mathbf{j}}$ and $\vec{\mathbf{B}} = 4\hat{\mathbf{i}} - 4\hat{\mathbf{j}}$, (b) $\vec{\mathbf{A}} = -2\hat{\mathbf{i}} + 4\hat{\mathbf{j}}$ and $\vec{\mathbf{B}} = 3\hat{\mathbf{i}} - 4\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$, and (c) $\vec{\mathbf{A}} = \hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and $\vec{\mathbf{B}} = 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$.

(a) $\vec{\mathbf{A}} = 3.00\hat{\mathbf{i}} - 2.00\hat{\mathbf{j}}$

$\vec{\mathbf{B}} = 4.00\hat{\mathbf{i}} - 4.00\hat{\mathbf{j}}$

$$\theta = \cos^{-1}\left(\frac{\vec{\mathbf{A}}\cdot\vec{\mathbf{B}}}{AB}\right) = \cos^{-1}\left(\frac{12.0 + 8.00}{\sqrt{13.0}\cdot\sqrt{32.0}}\right) = \boxed{11.3^\circ}$$

(b) $\vec{\mathbf{A}} = -2.00\hat{\mathbf{i}} + 4.00\hat{\mathbf{j}}$

$\vec{\mathbf{B}} = 3.00\hat{\mathbf{i}} - 4.00\hat{\mathbf{j}} + 2.00\hat{\mathbf{k}}$

$$\cos\theta = \left(\frac{\vec{\mathbf{A}}\cdot\vec{\mathbf{B}}}{AB}\right) = \frac{-6.00 - 16.0}{\sqrt{20.0}\cdot\sqrt{29.0}} \rightarrow \theta = \boxed{156^\circ}$$

(c) $\vec{\mathbf{A}} = \hat{\mathbf{i}} - 2.00\hat{\mathbf{j}} + 2.00\hat{\mathbf{k}}$

$\vec{\mathbf{B}} = 3.00\hat{\mathbf{j}} + 4.00\hat{\mathbf{k}}$

$$\theta = \cos^{-1}\left(\frac{\vec{\mathbf{A}}\cdot\vec{\mathbf{B}}}{AB}\right) = \cos^{-1}\left(\frac{-6.00 + 8.00}{\sqrt{9.00}\cdot\sqrt{25.0}}\right) = \boxed{82.3^\circ}$$

13. Let $\vec{\mathbf{B}} = 5.00$ m at 60.0°. Let the vector $\vec{\mathbf{C}}$ have the same magnitude as $\vec{\mathbf{A}}$ and a direction angle greater than that of $\vec{\mathbf{A}}$ by 25.0°. Let $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = 30.0$ m^2 and $\vec{\mathbf{B}} \cdot \vec{\mathbf{C}} = 35.0$ m^2. Find the magnitude and direction of $\vec{\mathbf{A}}$.

Let θ represent the angle between $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$. Turning by 25.0° makes the dot product larger, so the angle between $\vec{\mathbf{C}}$ and $\vec{\mathbf{B}}$ must be smaller. We call it $\theta - 25.0°$. Then we have

$$5A \cos\theta = 30 \quad \text{and} \quad 5A \cos(\theta - 25.0°) = 35$$

Then

$$A \cos\theta = 6 \quad \text{and} \quad A(\cos\theta \cos 25.0° + \sin\theta \sin 25.0°) = 7$$

Dividing,

$$\cos 25.0° + \tan\theta \sin 25.0° = 7/6$$

or

$$\tan\theta = (7/6 - \cos 25.0°)/\sin 25.0° = 0.616$$

Which gives $\theta = 31.6°$. Then the direction angle of A is

$$60.0° - 31.6° = 28.4°$$

Substituting back,

$$A \cos 31.6° = 6 \quad \text{so} \quad \vec{\mathbf{A}} = \boxed{7.05 \text{ m at } 28.4°}$$

Chapter 11.3: Vector Product

1. Given $\vec{\mathbf{M}} = 2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\vec{\mathbf{N}} = 4\hat{\mathbf{i}} + 5\hat{\mathbf{j}} - 2\hat{\mathbf{k}}$, calculate the vector product $\vec{\mathbf{M}} \times \vec{\mathbf{N}}$.

$$\vec{\mathbf{M}} \times \vec{\mathbf{N}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & -3 & 1 \\ 4 & 5 & -2 \end{vmatrix} = \hat{\mathbf{i}}(6-5) - \hat{\mathbf{j}}(-4-4) + \hat{\mathbf{k}}(10+12) = \boxed{\hat{\mathbf{i}} + 8.00\hat{\mathbf{j}} + 22.0\hat{\mathbf{k}}}$$

2. The displacement vectors 42.0 cm at 15.0° and 23.0 cm at 65.0° both start from the origin and form two sides of a parallelogram. Both angles are measured counterclockwise from the x axis. (a) Find the area of the parallelogram. (b) Find the length of its longer diagonal.

(a) $$\text{area} = \left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right| = AB\sin\theta = (42.0\text{ cm})(23.0\text{ cm})\sin(65.0° - 15.0°)$$

$$= \boxed{740\text{ cm}^2}$$

(b) The longer diagonal is equal to the sum of the two vectors.

$$\vec{\mathbf{A}} + \vec{\mathbf{B}} = [(42.0\text{ cm})\cos 15.0° + (23.0\text{ cm})\cos 65.0°]\hat{\mathbf{i}}$$

$$+[(42.0\text{ cm})]\sin 15.0° + (23.0\text{ cm})\sin 65.0°]\hat{\mathbf{j}}$$

$$\vec{\mathbf{A}} + \vec{\mathbf{B}} = (50.3\text{ cm})\hat{\mathbf{i}} + (31.7\text{ cm})\hat{\mathbf{j}}$$

$$\text{length} = \left|\vec{\mathbf{A}} + \vec{\mathbf{B}}\right| = \sqrt{(50.3\text{ cm})^2 + (31.7\text{ cm})^2} = \boxed{59.5\text{ cm}}$$

3. Two vectors are given by $\vec{\mathbf{A}} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}}$ and $\vec{\mathbf{B}} = -2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$.
M Find (a) $\vec{\mathbf{A}} \times \vec{\mathbf{B}}$ and (b) the angle between $\vec{\mathbf{A}}$ and $\vec{\mathbf{B}}$.

(a) We use the definition of the cross product and note that $\hat{\mathbf{i}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} \times \hat{\mathbf{j}} = 0$:

$$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = \left(1\hat{\mathbf{i}} + 2\hat{\mathbf{j}}\right) \times \left(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}}\right)$$

$$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = 2\hat{\mathbf{i}} \times \hat{\mathbf{i}} + 3\hat{\mathbf{i}} \times \hat{\mathbf{j}} - 4\hat{\mathbf{j}} \times \hat{\mathbf{i}} + 6\hat{\mathbf{j}} \times \hat{\mathbf{j}}$$

$$= 0 + 3\hat{\mathbf{k}} - 4\left(-\hat{\mathbf{k}}\right) + 0 = \boxed{7.00\hat{\mathbf{k}}}$$

(b) Since $\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right| = AB\sin\theta$, we have

$$\theta = \sin^{-1}\left(\frac{\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|}{AB}\right) = \sin^{-1}\left(\frac{7}{\sqrt{1^2 + 2^2}\sqrt{2^2 + 3^2}}\right) = \boxed{60.3^\circ}$$

6. Two vectors are given by these expressions: $\vec{\mathbf{A}} = -3\hat{\mathbf{i}} + 7\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$ and $\vec{\mathbf{B}} = 6\hat{\mathbf{i}} - 10\hat{\mathbf{j}} + 9\hat{\mathbf{k}}$. Evaluate the quantities (a) $\cos^{-1}[\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}/AB]$ and (b) $\sin^{-1}[|\vec{\mathbf{A}} \times \vec{\mathbf{B}}|/AB]$. (c) Which give(s) the angle between the vectors?

$$\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} = -3.00(6.00) + 7.00(-10.0) + (-4.00)(9.00)$$

$$= -124$$

$$AB = \sqrt{(-3.00)^2 + (7.00)^2 + (-4.00)^2} \cdot \sqrt{(6.00)^2 + (-10.0)^2 + (9.00)^2}$$

$$= 127$$

(a) $$\cos^{-1}\left(\frac{\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}}{AB}\right) = \cos^{-1}(-0.979) = \boxed{168^\circ}$$

(b) $$\vec{\mathbf{A}} \times \vec{\mathbf{B}} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -3.00 & 7.00 & -4.00 \\ 6.00 & -10.0 & 9.00 \end{vmatrix} = 23.0\hat{\mathbf{i}} + 3.00\hat{\mathbf{j}} - 12.0\hat{\mathbf{k}}$$

$$\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right| = \sqrt{(23.0)^2 + (3.00)^2 + (-12.0)^2} = 26.1$$

$$\sin^{-1}\left(\frac{\left|\vec{\mathbf{A}} \times \vec{\mathbf{B}}\right|}{AB}\right) = \sin^{-1}(0.206) = \boxed{11.9^\circ} \text{ or } 168^\circ$$

(c) Only the first method gives the angle between the vectors unambiguously because $\sin(180^\circ - \theta) = \sin\theta$ but $\cos(180^\circ - \theta) = -\cos\theta$; in other words, the vectors can only be at most 180° apart and using the second method cannot distinguish θ from $180^\circ - \theta$.

7. If $|\vec{A} \times \vec{B}| = \vec{A} \cdot \vec{B}$, what is the angle between $\vec{A}$ and $\vec{B}$?

We are given the condition $|\vec{A} \times \vec{B}| = \vec{A} \cdot \vec{B}$.

This says that $AB\sin\theta = AB\cos\theta$

so $\tan\theta = 1$

$\theta =$ $\boxed{45.0°}$ satisfies this condition.

10. A student claims that he has found a vector $\vec{\mathbf{A}}$ such that $(2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}) \times \vec{\mathbf{A}} = (4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}})$. (a) Do you believe this claim? (b) Explain why or why not.

(a) No.

(b) The cross-product vector must be perpendicular to both of the factors, so its dot product with either factor must be zero. To check:

$$\left(2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} + 4\hat{\mathbf{k}}\right) \cdot \left(4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - \hat{\mathbf{k}}\right) = \left(\hat{\mathbf{i}} \cdot \hat{\mathbf{i}}\right)8 + -9\left(\hat{\mathbf{j}} \cdot \hat{\mathbf{j}}\right) - 4\left(\hat{\mathbf{k}} \cdot \hat{\mathbf{k}}\right)$$

$$= 8 - 9 - 4 = -5$$

The answer is not zero.

No. The cross product could not work out that way.